



# FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to [grants@foundationfar.org](mailto:grants@foundationfar.org).

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

| Grant Information                                   |                                                                                  |
|-----------------------------------------------------|----------------------------------------------------------------------------------|
| Grant ID                                            | 22-000285                                                                        |
| Award Program                                       | Direct Fund/Non-Competitive                                                      |
| Project Title                                       | Field trial of gene-edited rice for drought tolerance                            |
| Grant Start Date                                    | 07/01/2022                                                                       |
| Grant End Date                                      | 06/30/2024                                                                       |
| Total Grant Budget                                  | \$50000                                                                          |
| Total FFAR Budget                                   | \$25000                                                                          |
| Matching Funder(s)                                  | Open Philanthropy Project (distributed through the FFAR award)                   |
| Final Report Date                                   | 08/30/2024                                                                       |
| Project Director/Principal Investigator Information |                                                                                  |
| Full Name                                           | Brian Staskawicz                                                                 |
| Email                                               | stask@berkeley.edu                                                               |
| Phone                                               |                                                                                  |
| Grantee Organization Information                    |                                                                                  |
| Organization Name                                   | The Regents of the University of California,<br>on behalf of its Berkeley Campus |
| Address                                             | 1608 Fourth Street, Suite 220                                                    |
| City, State Zip                                     | Berkeley, CA 94720                                                               |
| Tax ID                                              | 94-6002123                                                                       |
| Authorized Signing Official (ASO) Information       |                                                                                  |
| Name                                                | Jeffrey Bui                                                                      |
| ASO Title                                           | Contracts and Grants Officer, UCB, Sponsored Projects Office                     |
| Email                                               | jeffreybui@berkeley.edu                                                          |
| Phone                                               | 510 642-0120                                                                     |



## General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

Cali, Colombia

- 1.2. How many new jobs were created because of this grant funding?

None

- 1.3. How many jobs were maintained because of this grant funding?

None

- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

None

- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

None

## 2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Stomata are the primary sites of water loss in most plants and therefore serve as an excellent trait for adapting crops to current and future water limitations. This work sought to evaluate the in-field performance of two reduced stomatal density lines generated by CRISPR/Cas9. The genes targeted were EPFL10 and STOMAGEN, respectively. Both edited lines exhibited yield reductions in field conditions and signs of drought stress in water-limited conditions and well-watered conditions. The findings of this study indicate that stomatal density reductions mediated by editing the genes selected was ultimately not an effective mechanism to establish drought resilience in water-limited conditions. This work provides valuable insights into



stomatal reductions and agronomic performance and necessitates greater research to increase understanding of how stomata may be effectively tuned to address water limitations.

yield reductions in field conditions relevant to wild-type lines.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

The findings of this research are primarily relevant to researchers seeking to support the development of crops for use in agriculture. To date, there has been many studies seeking to generate stomatal density reductions towards greater water savings with limited field testing of lines generated. Our work provides novel insights regarding the agronomic outcomes of stomatal density reductions in a relevant field setting. While the work has identified that the current gene-edited lines produced are not yet ready for adoption into agricultural systems, the findings are still useful nonetheless. Fundamental insights generated by the field data collected indicate that additional research is required prior to this technological goal can be achieved. Future funding should be allocated to better understand how stomatal traits can be more effectively tuned towards effective agricultural outcomes.

- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:
  - a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
  - b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
  - c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

This project was able to establish the current state of the field in regards to stomatal density mediated improvements for water savings improvements in rice. The lines being tested were established using gene-editing targeting two separate genes which serve as positive regulators of stomatal density. The insights generated by this study were among the first study to comprehensively study the agronomic outcomes of this approach. Despite previous



efforts and investments in stomatal density mediated water savings, ours is among the first to study this approach in a field setting. Previous work using these gene edited lines in the greenhouse revealed promising physiological phenotypes which encouraged field trials undertaken. The discrepancy between greenhouse and field data indicates to future researchers in the space that greenhouse data may not fully represent the utility of traits in relevant agronomic settings. The findings of our study make a valuable contribution to those working to use genetic approaches to engineer stomata. They are a validation of the absolute necessity of field trials to understand any agronomic consequences of putative improvements. Future research must seek to understand how best to use genetic approaches for stomatal design and engineering that will lead to highly functional agronomic outcomes. Additional foci of future may include considerations of meaningful phenotypic assays in greenhouse conditions that serve as more relevant proxies of field performance, developmental and physiological engineering of stomata simultaneously, and identifying highly discrete targets for editing that may result in higher utility stomatal lines.

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

The major and primary goal of this project was to complete a singular field trial in well-watered and drought conditions of gene edited lines. This goal was achieved and successfully completed. The ultimate completion of the study was delayed on the basis challenges with importation of seeds at the collaborating institution. Data were collected on or before March 5, 2024.

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

No

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)



The singular and specific objective of this grant funding was to determine in well-watered and water-limited conditions, the agronomic properties of gene-edited lines that exhibited variation in stomatal density. This outcome was achieved in collaboration with CIAT an international research organization present in Cali, Colombia. Results generated indicated that gene-edited lines exhibited lower yield overall, were less thermotolerant, and broadly were not suitable for cultivation. While these findings signify that the tested technology is not ready for deployment, they do support the establishment of more focused research programs. Many research groups are pursuing stomatal mediated improvements for water-savings and climate adaptation. Great opportunity still exists to use stomata as a mechanism to adapt current germplasm to emerging climate threats. The results of our study suggest that pleiotropy of the gene targeted may have been at least partially responsible for the sub-optimal agronomic outcomes. Future work pursuing gene editing approaches must pursue additional gene targets that are less likely to exhibit pleiotropic effects. Future research programs may also benefit from engineering physiological adaptations simultaneous with novel developmental regimes in order to account for some of the detected negative effects of developmental program edits. Broadly, this work has indicated that gene-editing mediated stomatal density adjustments are not the field. Additional approaches both in terms of genes targeted and mechanism of improvement must be explored towards generating climate change adapted germplasm currently suitable for use in. Our work is the gene-edited field trial of germplasm generated at UC Berkeley and one of the earliest attempts at using gene editing in field conditions. While the outcome of the trials did not result in technology fit for immediate use, the insights generated are valuable. Furthermore, this grant funding allowed for the establishment of valuable partnership between the Innovative Genomics Institute and CIAT.

- 2.7. Discuss efforts taken to ensure the approach is scientifically rigorous.  
Include approaches taken to ensure robust and unbiased results.

Our work was conducted with a collaborator who specializes in field trialing germplasm in rainout shelters. They have extensive experience and equipment towards achieving robust data collection. Genotypes were randomly distributed through the plot to account for environmental variation. Genotype names were coded, so those conducting the research would be blind to the germplasm being tested and would not contribute any bias. A drone was used to gather chlorophyll SPAD levels and thermal images. Automated image processing was used to process these images to avoid bias. Randomization, blindness, and automation built into the experimental design ensured that data collected was both robust and unbiased.

- 2.8. List key stakeholders that could be served by results of this project.



The primary stakeholders of this research based on the results of the project would be research communities seeking to establish mechanisms for improved drought tolerance or water-use efficiency in crops.

- 2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

Nothing to report

- 2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

A major delay ensued from a permitting error that occurred with regards to import of seeds into the Colombia. The permit initially issued was not accepted by Colombia, and our seed materials were incinerated. Reissuance of the corrected permit was a slow process and resulted in a significant delay. Working closely with our collaborators in Colombia we were able to ensure that the correct permits were issued and that the seeds were successfully imported.

- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

A postdoctoral trainee was able to lead and execute the work on this project furthering his development as a principal investigator and research executive.



- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

Nicholas Karavolias, Postdoctoral scholar, was the primary investigator. Additional technicians at our collaborating institution were also involved. This included researchers Michael Gomez Selvaraj and Diego Guzman.

### 3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

Nothing to report

- 3.2. Please provide a list of citations for the information products produced during the project.

Nothing to report

- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

Nothing to report

- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

Nothing to report

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).



|            | # Filed (Enter Numeric Value) | # Approved (Enter Numeric Value) | Patent Numbers (Separated by commas) |
|------------|-------------------------------|----------------------------------|--------------------------------------|
| Patents    | #                             | #                                |                                      |
| Copyrights | #                             | #                                |                                      |
| Trademarks | #                             | #                                |                                      |
| Inventions | #                             | #                                |                                      |

3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

Nothing to report

3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that others can access and re-use these data in the future?

Details of the work will be communicated within the research community

## 4. Data Management

4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award.

Nothing to report

4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

Nothing to report



- 4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

Nothing to report



## Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Brian Staskawicz

PI Signature: DocuSigned by:  
Brian Staskawicz 96512AF2662C46C... Date: 9/3/2024

Authorized Signing Official (ASO) Name: Jeffrey Bui

ASO Signature: DocuSigned by:  
Jeffrey Bui B9FAC1D214454E3... Date: 9/3/2024